Catherine Yousry Sobhy Khalaf

Computer And Information Sciences Student – Ain Shams University
(+20)1227908616 | Al-Khosous, Qalyubia Governorate, Egypt.
catherine.yousry.dev@gmail.com
LinkedIn | GitHub

Objective:

Detail-oriented Junior Data Analyst and Computer Science student skilled in SQL, Python, Microsoft Excel, and Power BI dashboard development. Experienced in ETL processes, exploratory data analysis (EDA), and machine learning pipelines to transform complex datasets into actionable business insights.

Education:

Ain Shams University, Faculty of Computer and Information Sciences Sep 2024 – Present

- Bachelor of Computer and Information Sciences (Major: Mainstream)
- Expected Graduation Year: 2028

Work & Internship Experience:

Machine Learning Trainee | National Telecommunication Institute (NTI) (Aug 2026)

- Completed an intensive training program in collaboration with ITIDA, focusing on end-to-end model development and advanced data processing.
- Built a **Credit Card Fraud Detection Pipeline** addressing severe class imbalance using **SMOTE** and deploying predictive models via **Streamlit**.
- Applied both **Supervised and Unsupervised Learning** techniques, evaluating algorithms like **Random Forest** and **XGBoost** to optimize precision and recall metrics.

Technical Projects:

- **Credit Card Fraud Detection Pipeline** (Sep 2026)
  - Conducted exploratory data analysis (EDA) on imbalanced financial transactions to identify fraud indicators.
  - Handled data sampling with SMOTE, trained ensemble ML models, and deployed an interactive web dashboard using Streamlit.
- **Retail Sales & Business Intelligence Dashboard** (Aug 2026)
  - Cleaned and validated multi-channel sales data using SQL queries and engineered interactive Power BI dashboards to track daily business trends and KPI metrics.
- **E-Commerce Analytics Suite & Executive Dashboard** (Jul 2026)
  - Developed an end-to-end Python data analysis pipeline to process, clean, and transform transactional retail datasets.
  - Implemented automated exploratory data analysis (EDA) including price outlier trimming, missing data filtering, and sales tier segmentation.
  - Designed executive analytical dashboards displaying key metrics such as monthly revenue trajectories, top international markets, and transaction peak hours using Seaborn & Matplotlib.
- **LIP.AI – Loan Status Prediction Platform** (May 2025 – June 2025)
  - Developed an end-to-end machine learning web application to evaluate applicant financial profiles and predict loan approval status.
  - Trained and tuned classification models using Python, and deployed an interactive user interface using **Streamlit**.

Activities & Leadership:

- **Front-End Committee Leadership | MSP, Ain Shams University** (Sept 2025 – Dec 2025)
  - Coordinated committee workflows, managed task follow-ups, and prepared technical training materials.
  - Provided practical support during sessions and evaluated members' assignments to ensure learning quality.

Certifications & Training:

- **Full-Stack Web Development Program** – Sprints (June 2026)
- **Programming using Python** – Sprints & Microsoft (2025)
- **Web Development** – Sprints & Microsoft (2025)

Skills:

- **Technical Skills:**
  - **Data Analytics & BI:** Power BI, Microsoft Excel, Exploratory Data Analysis (EDA), Feature Engineering, Data Cleaning, KPI Dashboards.
  - **Programming & Libraries:** Python (Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn), SQL, C++, Java.
  - **Machine Learning:** Supervised & Unsupervised Learning, Imbalanced Data (SMOTE), Predictive Modeling.
  - **Databases & Tools:** PostgreSQL, SQLite, Streamlit, Git/GitHub, Microsoft Office.
- **Soft Skills:**
  - Analytical Thinking, Problem Solving, Teamwork, Communication Skills, Presentation Skills, Time Management.
- **Languages:** Arabic (Native), English (Very Good).